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Risk-Based Completion Cost Overrun Ratio Estimation in Construction Projects Using Machine Learning Classification Algorithms: A Case Study

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Abstract: Estimating the completion cost accurately in the early phases of construction projects is critical to their success. However, cost overruns are almost inevitable due to the risks inherent in construction projects. Hence, the completion cost fluctuates throughout the execution phase and requires periodic updates. There is a need for a prompt and user-friendly completion cost estimation model that accounts for fluctuating risk scores and their impacts on the total cost during the execution phase. Machine learning (ML) techniques could address these requirements by providing effective methods for tackling dynamic systems. The proposed approach aims to predict the cost overrun ratio classes of the completion cost according to the changes in the total risk scores at any time of the project. Six classification algorithms were utilized and validated by employing 110 data points from a globally operating construction company. The performances of the algorithms were evaluated with validation and performance indices. The decision tree classifier surpassed other algorithms. Although there are some research limitations, including risk perception, data gathering restrictions, and selecting proper ML algorithms upon data properties, this research improves the planning abilities of construction executives by providing a cost overrun ratio based on changing total risk scores, facilitating swift and simple assessments at any stage of a construction project's execution.

Keywords: classification; cost estimation; machine learning; cost overrun; case study; risk score

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1. Introduction

Cost estimation is a significant indicator of the success of construction projects [1]. However, estimating the cost of construction is a risky aspect of the process due to the dynamic nature of the construction industry. Most construction projects exceed their estimated cost for various reasons, as demonstrated by multiple studies [2–6].

The estimated completion cost of the project incorporates the anticipated risks associated with the construction activities [7–9]. Moreover, the ISO 9001:2015 Quality Management System requires implementing a "Risk management" approach in all business processes [10]. Generally, in the early stages of the project, the executives commenced identifying and assessing potential risks. These risks are defined and evaluated as the project's risk register [11]. These registers delineate the potential risks and their associated likelihood and impact estimates as determined by the executives [12]. Moreover, they are the primary tool of risk management utilized to detect potential risks within a project or organization, facilitating methodical knowledge management. However, the probability and impact of risks evolve during the project execution phase, and the completion cost overrun ratios also fluctuate based on these changing risk scores. Furthermore, as the project progresses, certain risks arise while others dissipate, and it is challenging to foresee the potential cost overrun effects of these risks individually. It is essential to recognize that risk registers include unexpected cases not addressed in contract clauses and excluded from cost estimation procedures. Consequently, it is essential and beneficial to develop an approach for estimating risk-based cost overruns throughout the execution phase of

construction projects to achieve successful cost estimation. The proposed approach should be practical and capable of producing results quickly due to the high workflow speed in the execution period of a construction project. Since the estimations are mainly conducted monthly and within a limited duration, the model must be adaptive and integrate new data that accumulates periodically. This user-friendly approach can ensure executives evaluate the potential cost overrun ratios at any stage of the project. Moreover, they can develop a mitigation action plan depending on these estimated ratios. ML approaches offer significant benefits by facilitating data processing, generating comprehensible outcomes, and elucidating the importance of various features [13–17]. Moreover, these techniques are widely used in construction management to forecast performance [18], productivity [19], safety issues [20–23], and cost estimation [24]. In this study, the emphasis was not placed on causality, which statistical approaches might elucidate. Instead, the main focus was prediction, leading to the preference for ML algorithms. These algorithms have inherent advantages since they can surpass the limitations of explicit model structures and suppositions about distributions of data frequencies, continuously enhancing their performance via experience and maximizing their predictive capabilities [15].

Additionally, although extensive research on cost estimation during the tender or preliminary stages is highly beneficial, there is a need for an approach applicable to the execution phase that can be simply and promptly updated at any moment. Furthermore, assessing the cost overrun ratio based on the total cost and risk scores, rather than breakdown components, facilitates the estimation process in the execution phase of a construction project.

The main objective of this research is to provide a systematic approach for determining the construction cost overrun ratio class by considering the assessment of the total risk score throughout the execution phase. In this context, a proposed ML-based classification approach was implemented on construction projects obtained from a globally operating construction company for practical application.

2. Literature Review

2.1. Cost Overrun in Construction Projects

Cost overruns frequently occur in construction projects. Several research studies have focused on the causes of cost overruns. Mansfield et al. [25] conducted a questionnaire on Nigerian construction projects. They determined that finance, contract management, fluctuating site conditions, and material shortages were the primary factors driving cost overruns. Another questionnaire was performed in Indonesia about cost overruns on high-rise projects. Results indicate that increases in material costs, project complexity, and poor estimation were the primary factors contributing to cost overruns [26]. Frimpong et al. [27] used a questionnaire to investigate groundwater construction projects in Ghana in terms of delays and cost overruns. The results reveal that payment difficulties and poor contractor management were critical issues for these concerns. Bin Seddeeq et al. [28] focused on cost overruns in the Oil and Gas Construction Industry in Saudi Arabia. The survey results indicate that design and scope changes, inadequate planning, errors in design, unclear scope comprehension, and underestimation of costs and schedules are the primary causes of time and cost overruns. Alshihri et al. [4] conducted a comparable study on building projects in the same region. They determined that change orders and selecting the lowest bidder are the main factors contributing to cost overruns. Xie et al. [29] performed a questionnaire about the same topic in China and found that market price and currency fluctuations, insufficient risk and contract management, design modifications, and constraints related to project locations were revealed as the most significant factors.

Moreover, numerous studies have developed a strategy for managing cost overruns in construction projects. Dikmen et al. [30] developed a fuzzy risk assessment methodology to identify cost overrun risks in international construction projects. They identified and modeled risks and utilized fuzzy rules to produce a final risk rating in their methodology. Using random forest analysis, Shoar et al. [31] assessed the cost overrun for high-rise resi-

dential structures. Based on the comparative analysis of prediction performance, random forests outperformed support vector regression and multiple linear regression. Chadee et al. [6] employed a sequential mixed-methods strategy, using reference class forecasting to improve the precision of cost estimation in public sector social housing projects, and their model effectively enhances accuracy in forecasting cost overruns. Alhammadi et al. [5] utilized the structural equation model to investigate the correlations between the causes of cost overruns and their mitigation strategies. Inadequate contract management and inaccurate cost estimation were the primary factors contributing to cost overruns. They recommended prioritizing mitigation strategies to reduce cost overruns. Hussain et al. [32] conducted a study on reducing cost overruns in rail projects by investigating the 5D-BIM framework. Results indicate that implementing the 5D-BIM framework mitigates cost overruns by delivering accurate cost estimations and dynamic financial control throughout the project lifecycle. Finally, Sohrabi and Noorzai [9] examined cost overrun estimation for water-related projects utilizing case-based reasoning based on varied risks and uncertainties in ML models. Their model has demonstrated a favorable correlation with ML algorithms for predicting cost overruns.

According to the examined literature, contract management, financial concerns, and risks associated with construction sites and materials are significant factors contributing to cost overruns in construction projects. Although these factors are crucial concerning cost overrun, there is a need to implement risks not covered in contract clauses or not included in cost estimation processes. Changes in the scope of work or arising risks related to site conditions are mainly included in cost estimation processes when they are identified. However, other risks were periodically reported into risk registers from all departments to project executives as required by ISO 9001:2015, and the effect of this risk score is mainly not reflected in the estimated completion cost [10]. Therefore, in the framework of this study, a completion cost overrun ratio estimation approach is proposed based on the risk score reported in risk registers.

2.2. Methods to Estimate Construction Costs

Numerous methodologies exist in the literature for estimating construction costs. Certain studies focus on particular construction costs rather than the total completion cost. Chang and Shinozuka [33] calculated the life-cycle cost of bridge construction by aggregating both anticipated and potential unforeseen expenses by users. Adeli and Wu [34] predicted the costs associated with reinforced concrete pavements in highway projects using a neural network. Ma et al. [35] and Lee et al. [36] utilized building information modeling (BIM) for cost estimation to determine the concrete costs of a six-story building and the tiling costs in international construction projects, respectively. Gurcanli et al. [8] assessed the safety costs of residential buildings during the initial phases of the project utilizing activity-based costing.

Moreover, numerous methods exist to estimate the total cost of the construction projects. The researchers mainly used neural networks [1,16,37–46], different types of regression models [1,31,42,47–51], case-based reasoning [1,52–55], BIM-based cost estimation [56–59], support vector machine and random forest [31,60–62], genetic algorithm [37,38,52], and Monte Carlo simulation [63,64] to estimate the total construction cost of a project, mainly in early phases. Recent studies indicate that ML approaches have been progressively employed in the construction industry for cost estimation. Alshboul et al. [45] employed ML algorithms, including extreme gradient boosting, deep neural network, and random forest, to estimate green building costs. In the research, extreme gradient boosting gave the highest accuracy, followed by deep neural network and random forest. Park et al. [62] employed a combination of algorithms, including random forest, support vector machine, and CatBoosting, to estimate the total cost of building construction projects in the early phases. Their suggested model demonstrated superior predictive performance compared with the individual models. Park and Yun [59] studied the total cost of a building project in the design phase, utilizing a BIM model and ML. They determined that precise construction cost estimation

is achievable by integrating the general characteristics of the building with BIM attributes. Furthermore, Das et al. [16] employed ML methods, including artificial neural networks, natural gradient boosting, and linear regression, to forecast construction costs, with natural gradient boosting exhibiting superior performance in predicting the total construction cost.

Additionally, it is noteworthy that most of the research focused on the early phases of the construction process. Neural networks [1,37,38,40,42,65], regression models [1,31,42,49,51], and BIM-based cost estimation [57,59] were predominantly employed methods mainly in residential structures and high-rise projects to estimate total costs during preliminary stages. Although the significant value of these studies in guiding future research, there is a need for estimating models that can be updated efficiently and swiftly at any moment in the execution phase of a construction project. Estimating the completion cost of a construction project is crucial not just in the design, tender, or commencement phase but also during the execution phase. While most of the project's risks identified in contract conditions were initially included in the completion cost, unexpected risks are regularly assessed in risk registers during the project's execution phase and should be reflected in the total completion cost. There is limited research on estimating total completion cost through the execution phase of a construction project. Cheng and Hoang [60] researched the integration of ML methods to achieve precise cost estimation and measurement of uncertainty. Their model could be applicable in the execution phase and showed higher performance in estimating compared with conventional methods, with a high level of reliability and usability in construction management. In addition, Mhady et al. [65] conducted a study on fuzzy inference systems and artificial neural networks, utilizing data from residential projects to accurately estimate costs during the early and late phases of construction. There is a requirement for more approaches that allow contractors to foresee the final cost of completion promptly while executing the projects. ML is highly valuable in this area due to its demonstrated effectiveness and integration of many approaches. Within the scope of this study, ML classification algorithms were employed to efficiently and rapidly update the cost overrun ratios of total cost based on the total risk scores at any point during project execution.

3. Methodology of the Proposed Approach

The employed methodology of this study to achieve the objectives is composed of four main stages: (1) data gathering, (2) data preprocessing, (3) model selecting and modeling, and (4) evaluating and validating the model.

In the first stage of the methodology, the risk registers and expected completion cost data of the construction projects are collected. Following the project's initiation, regular risk assessment and estimates of the completion cost of the project period begin. Data regarding these files throughout the same timeframe are gathered. To accomplish the goal of making this study easier to apply, it is suggested to gather both the total cost and total risk score of the project instead of breakdowns. The total risk score of the project is calculated depending on the probability and impact of the risk items. The evaluation of these factors might differ depending on the projects' assessment methods. Based on the project characteristics, some subcategories of risks may exist, and the executives could assign different weights to these categories. The total risk score computation is presented in Equation (1):

$$Total\ Risk\ Score = \frac{1}{k}\sum_{e=1}^{k}\left(\sum_{i=1}^{m}\left(W_{i,e}\frac{1}{n_i}\sum_{j=1}^{n_i}\left(P_{i,j,e}\times I_{i,j,e}\right)\right)\right) \quad (1)$$

where $P_{i,j,e}$ is the probability value assigned by the eth executive for j^{th} risk item in i^{th} risk category, $I_{i,j,e}$ is the impact value assigned by the eth executive for j^{th} risk item in i^{th} risk category, n is the total number of risk items, $W_{i,e}$ is the weight value assigned by the eth executive for i^{th} risk category, m is the total number of categories, and k is the total number of executives.

Depending on the specific project types, different approaches can be employed to provide risk register reports and completion cost assessments for construction projects. Hence, it is recommended to assess projects using a classification system based on their characteristics. The recommended project types are listed in Table 1.

Table 1. Project Types and Groups [66].

Code	Project Type (PT)
PT.01	Residential Housing Projects
PT.02	Heavy Civil Projects and Infrastructure
PT.03	Commercial and Institutional Projects
PT.04	Specialized Industrial Projects

In the second stage, data are preprocessed. The collected risk records and completion cost data of the projects are consolidated. Although the currency and magnitude of the project cost might vary depending on the size and location of the project, using the cost overrun ratio presented in Equation (2) rather than the total cost value in calculations, eliminates the differences.

$$TC_i\% = \frac{(TC_i - TC_{i-1})}{TC_{i-1}} \times 100 \quad (2)$$

where TC_i is the total cost of ith prediction period of the project.

The suggested approach involves classification algorithms to categorize the data into several classes. These classifications are contingent upon the project's requirements and characteristics. It is recommended that guidance be sought from the project executives. These classes should be set out to facilitate the presentation for executives or project managers about project total cost escalation depending on total risk scores and offer enhanced managerial understanding and advantages.

Afterward, the input dataset is randomly divided into training and testing datasets, with two-thirds assigned to the training dataset and one-third assigned to the testing dataset. The training set is used to improve the model. On the other hand, the testing set is chosen randomly and kept separate to assess the model's performance on data that it has not been exposed to before.

In the third stage, algorithms are selected for classification and subsequent modeling. The primary objective of the suggested approach is to categorize the cost overrun ratio of project completion cost based on total risk scores, depending on the determined classes. It is recommended to utilize supervised learning. Supervised learning entails linking each observation with a predictor observation and a corresponding response measurement. A suitable model should be able to anticipate future observations or infer the correlation between the response and predictor variables using observed data [67]. The choice of algorithm is determined by both the sample size and the characteristics of the data. Furthermore, it is acknowledged that no definitive ML technique is generally optimal for resolving specific problems [68].

In the fourth stage of the methodology, the employed algorithms are evaluated and validated by comparing the expected and actual class outcomes throughout training and testing. The models are assessed by calculating the confusion matrix and accuracy. Once the algorithms with an accuracy level exceeding 70% are chosen, performance indices are computed to overcome the accuracy paradox. The algorithms are ranked based on their performance metrics. Subsequently, resampling validation methods are employed to examine the testing performance of models, and the accuracy is computed for each subsampling. Performance indices are produced for these algorithms if the average accuracy exceeds 80%. The algorithm with the highest accuracy and performance indices is chosen from these options, and its parameters are optimized.

The employed methodology of the research is presented in Figure 1.

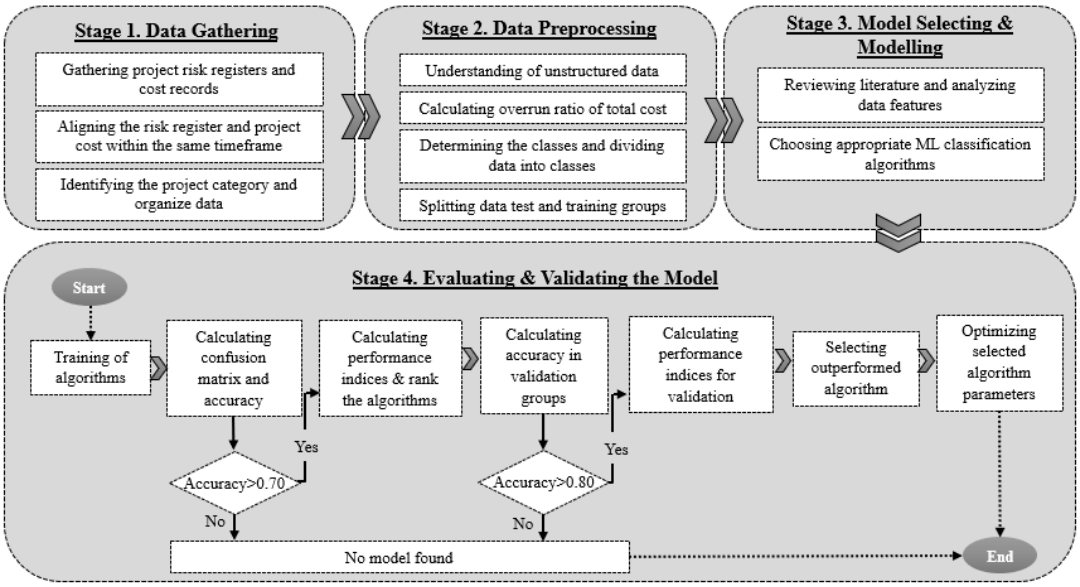


Figure 1. Methodology of the proposed approach.

4. Implementation of the Proposed Approach: A Case Study

Information about overall expenses and evaluations of potential risks were gathered from a general contractor that predominantly undertakes projects on a global scale and specializes mainly in constructing airports and high-rise buildings. A case study was conducted using this data to illustrate how the proposed approach can be applied in practice.

4.1. Stage 1: Data Gathering

The risk register and completion cost of projects were collected from the analyzed construction company. Various risk registers were created to meet the specific requirements of the client and project. All risk registers were standardized by integrating them into an identical format. To mitigate data limitations, a comprehensive review of all registers was conducted. To minimize risk perception and standardize the assessment of risk scores uniformly inside an organization, the probability and impact scales presented in Tables 2 and 3 were employed. The executives of the data-gathered company recommended these scales for the provided data.

Table 2. Risk probability scale.

Score	Probability Definition
5: Very High Probability	>70%
4: High Probability	51–70%
3: Probable	31–50%
2: Low Probability	10–30%
1: Very Low Probability	<10%

Table 3. Risk impact scale.

Score	Expected Financial Impact	Execution Impact
5: Very High Impact	>3% contract value	Suspension of execution and/or termination
4: High Impact	1.8–3.0% contract value	Major interruption on execution
3: Medium Impact	1.2–1.8% contract value	Serious interruption on execution
2: Low Impact	0.6–1.2% contract value	Minor interruption on execution
1: Very Low Impact	<0.6% contract value	Insignificant interruption on execution

As proposed in the methodology, the cost overrun ratios of each estimation in projects were calculated. Depending on the guidance of executives of the company from which

data were gathered, four classes were used for cost overrun ratios, as illustrated in Table 4. Class 0 represents the non-changed completion cost in the estimation. Therefore, this circumstance might be considered negligible, requiring no action. Class 1 comprises minor cost increases depending on total risk scores, while Class 2 indicates significant increases, and Class 3 signifies severe increases in the completion cost depending on total risk scores and demands mitigation actions depending on reported risks.

Table 4. Classes for cost overrun ratios.

Total Cost (TC) Increase %	Class
TC% $\leq$ 0%	0
0% < TC% < 5%	1
5% $\leq$ TC% < 10%	2
10% $\leq$ TC%	3

Then, the gathered risk registers were matched with the estimated project completion cost within the same timeframe. Finally, the projects' total risk scores and expected completion costs were classified based on their various project types, as presented in Figure 2.

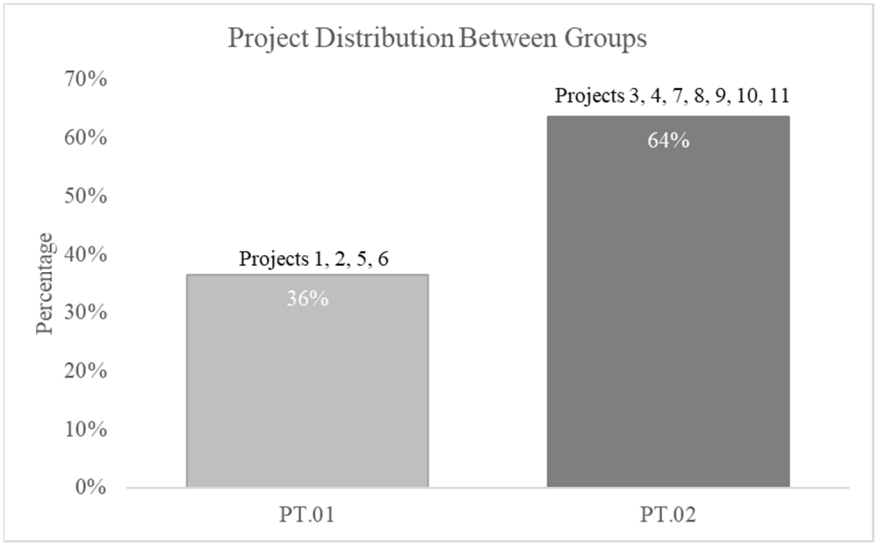


Figure 2. Groups depending on the project types.

4.2. Stage 2: Data Preprocessing

The PT.01 group consists of four residential projects, which have a total of 50 data points. On the other hand, the PT.02 group comprises seven heavy civil projects, primarily focused on airport-related projects, and has a total of 60 data points. Considering the cost data varies based on the projects, these data points were scaled using Equation (2). Subsequently, the data points were classified into four class labels, as recommended in Table 4. In the PT.01 group, 51% of the data points corresponded to Class 0, 35% to Class 1, 10% to Class 2, and 4% to Class 3. In addition, the PT.02 group had 47% of the data points in Class 0, 36% in Class 1, 10% in Class 2, and 7% in Class 3. While the distribution of data points between classes may not be uniform, these classes offer valuable perspectives on management. For a few projects, the estimated completion cost is expected to exceed the previous estimation by 10%. Consequently, there is an anticipation that the frequency of data will decline as the total cost overrun ratio rises. The development of classification algorithms and the acquisition of findings were achieved using Python 3.11 programming, a robust computational tool capable of effectively implementing many ML approaches. Following the interpretation, data preparation, and library loading in Python, the data

frame was converted into an array. Afterward, the dataset was divided into training and test sets. Training purposes were assigned two-thirds of the dataset, while one-third was set aside for validation.

4.3. Stage 3: Model Selecting and Modeling

The main objective of this study is to forecast the cost overrun ratio class of the project's total cost depending on total risk scores. ML categorization techniques were utilized due to the nature of the problem, and supervised learning was implemented depending on the features of the data. This technique entails utilizing existing data to acquire knowledge on how to allocate new data records to certain output classes, and it is highly efficient in predicting analytics of data [15]. Additionally, classification strategies can handle intricate factors and their efficacy in managing categorical variables [69].

Several studies in construction management have employed ML categorization algorithms to forecast performance, productivity, and safety issues. Wong [18] employed logistic regression for classification by examining 68 construction projects in England to forecast contractor performance using tender evaluation criteria of clients. Two categories indicated contractors' performance, and the logistic regression classifier model accurately predicted contractor performance in 75% of instances. Desai and Joshi [19] employed the decision tree algorithm to examine labor productivity in construction projects. The findings revealed that the decision tree algorithm yielded greater accuracy than conventional labor productivity approaches. Moreover, Mistikoglu et al. [20] employed the decision tree algorithm to examine fall incidents. They applied a categorization approach with distinct classes, each representing a cause of fall accidents, and they concluded that the decision tree classifier achieved a high level of accuracy in classifying incidents of roofers falling. Further research has been conducted on safety concerns utilizing ML categorization approaches. Drozd [21] utilized logistic regression to forecast the binary classification of accidents, determining whether they occurred. This was performed by considering a range of independent factors associated with building site safety measures and conditions. A total of 339 observations were analyzed, and the logistic regression model identified significant predictors of accidents. Choi et al. [17] applied logistic regression, decision tree, random forest, and AdaBoost algorithms to design a prediction model for fatal accidents in the construction industry. The study aimed to forecast the likelihood of deadly accidents by categorizing them into "general accidents" and "fatal accidents". Based on the findings, the random forest model had the highest level of accuracy in predicting outcomes for classes. Zhang et al. [70] employed the random forest classifier to discover significant factors contributing to work delays and affecting decision-making. Moreover, Scott and Bommu [24] utilized decision tree and random forest algorithms to ascertain the presence of cost overruns in construction projects. These ML models exhibit a high degree of accuracy in predicting cost overruns. It is known that there is no ML algorithm that has been proven to be the most effective for addressing specific problems [68]. Hence, it is essential to assess the efficacy of the algorithms using various approaches. The following ML classification algorithms were utilized due to their frequent usage, user-friendly nature, and ability to evaluate performance:

- Logistic Regression for Classification (LRC)
  The LRC algorithm is a classification approach that utilizes logistic functions to categorize data points based on their numerical values. The sigmoid function is employed to provide a clear distinction between different groups.
- K-Nearest Neighbors (KNN)
  The KNN methodology is a simple and resilient computing tool. This algorithm is a highly efficient technique for dividing non-linear data into distinct segments. This classifier employs multiple metric methods. The research utilized the default "Minkowski" approach with a 2nd degree, which gives Euclidian distance. For the closest neighbor number in the function, there are different results due to the operational logic of the function. Within the scope of this study, the value is set to 1 in order

to simplify the classification process by choosing the closest neighboring data point. This option facilitates a more straightforward decision-making process to enhance precision while also potentially decreasing computing complexity and improving the speed of categorization [71].

- Support Vector Classifier (SVC)
  The objective of this function is to identify the line that provides the maximum margin between groups. SVC enables the differentiation between binary and multi-class classification on a dataset using a Kernel function, allowing for both linear and non-linear separation [72].
- Naïve Bayes Classifier (NBC)
  The Naïve Bayes classifier operates by utilizing the principle of conditional probability. NBC minimizes test error by assigning each observation to the class that has the highest probability, determined by its predictor values [72].
- Decision Tree Classifier (DTC)
  DTC works based on the majority decision principle. This algorithm focuses on how independent factors divide data records into class labels by analyzing the training data [15]. Subsequently, the algorithm generates a classifier represented as a hierarchical structure, like a tree, composed of nodes, branches, and leaves. The root node symbolizes the complete dataset. Afterward, the data are divided according to a specific characteristic, resulting in the creation of subnodes. These nodes are linked by branches, which correspond to the associated variables. Leaves, on the other hand, reflect the final determined class labels. DTC has the benefit of comprehending and explaining complicated problems in an understandable hierarchical structure [17].
- Random Forest Classifier (RFC)
  The RFC method is a widely used classification algorithm that utilizes majority voting and integrates many decision tree principles. The act of creating an ensemble of trees and allowing them to vote for the most prevalent class has led to notable enhancements in classification accuracy [73].

4.4. Stage 4: Evaluating and Validating the Model

The algorithms were trained using two-thirds of the data, employing LRC, KNN, SVC, NBC, DTC, and RFC. After obtaining the models based on the training group, predictions were generated for the remaining one-third of the data. The total number of data sets is 110, including 50 from the PT.01 group and 60 from the PT.02 group. Sixteen data points from the PT.01 group and twenty data points from the PT.02 group were employed for validation. This section aims to verify the accuracy of the models' predictive capabilities and evaluate their efficiency. The Confusion Matrix is a widely used tool in the ML field for evaluating classification algorithms' performance. The rows of the confusion matrix show the number of actual classifications, and the columns reflect the number of predicted classifications. The diagonal of the table shows the correct classified data, namely accurate predictions. The confusion matrices in heatmap format for the six classifiers are shown in Figures 3 and 4 for the PT.01 and PT.02 groups, respectively. The intensity of the color corresponds to the concentration of data within the cell, and these tables match the suggested classes described in Table 4.

Subsequently, the classification accuracies (ACC) of each algorithm within the groups were computed using Equation (3) where CC is correct classifications and TS is the total samples, and the outcomes are displayed in Table 5.

$$ACC = \frac{CC}{TS} \tag{3}$$

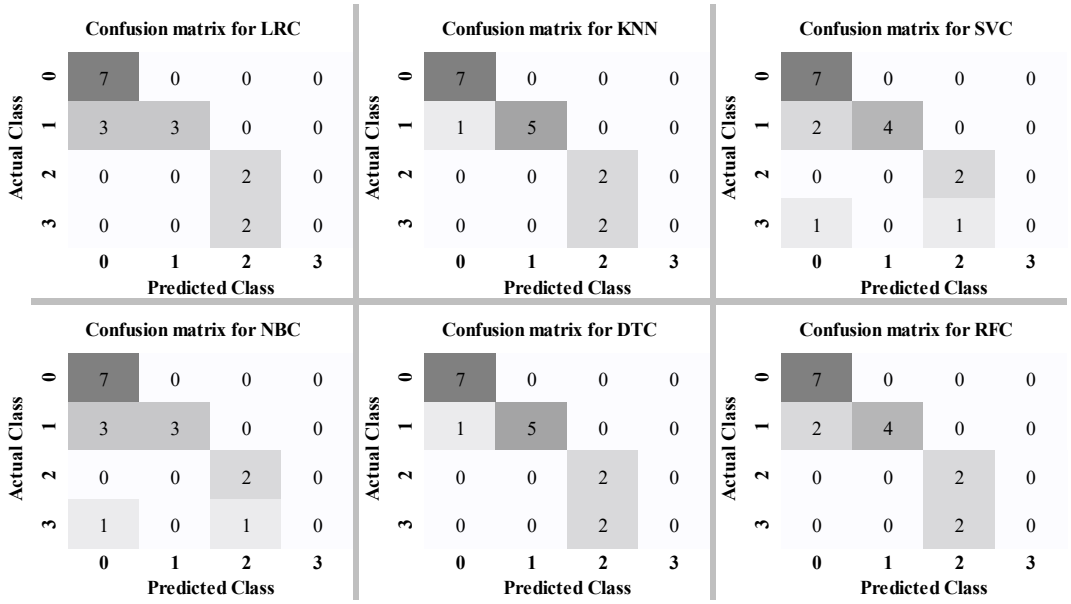


Figure 3. Heatmap of confusion matrices for the PT.01 group.

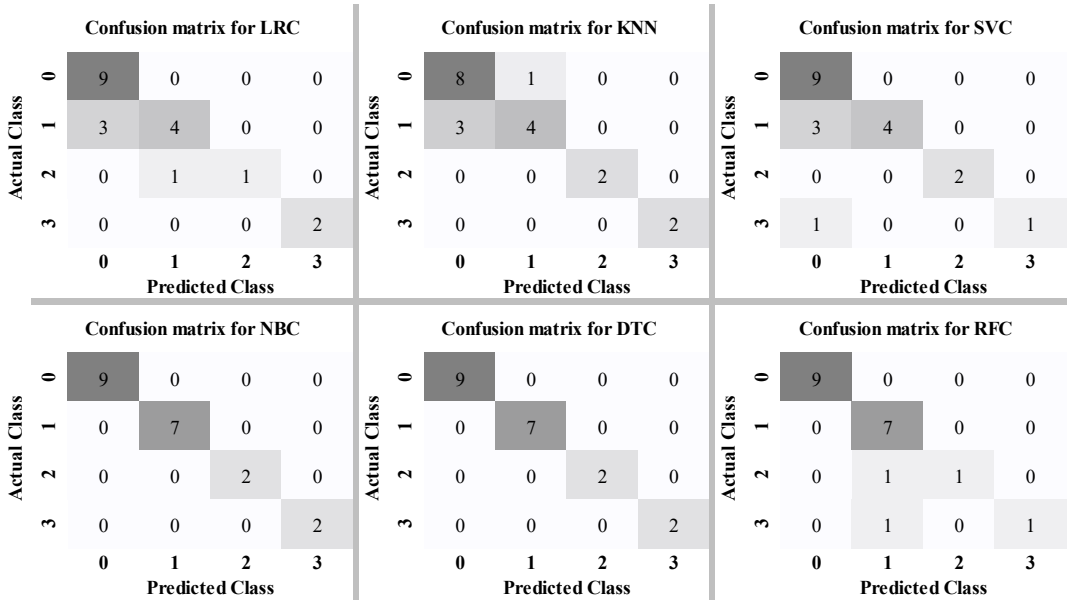


Figure 4. Heatmap of confusion matrices for the PT.02 group.

Table 5. Accuracy percentages of PT groups according to algorithms.

Algorithms	PT.01	PT.02
LRC	0.71	0.80
KNN	0.82	0.80
SVC	0.76	0.80
NBC	0.71	1.00
DTC	0.82	1.00
RFC	0.76	0.90

All of the algorithms achieved an accuracy level beyond 70.00%. The accuracy level of all algorithms exceeded 70% limit, only the performance of LRC and NBC in PT.01 group demonstrates a degree of accuracy that is nearly sufficient.

Nevertheless, it is important to note that larger accuracy percentages may not always reflect a high level of classifier performance, as indicated by the accuracy paradox [74]. The accuracy computation is derived from the summation of the diagonal elements of the confusion matrix across all data points. However, if the algorithm categorizes all the data into a single class and if the data are imbalanced, meaning there is a large number of data points in one class compared with others, the accuracy ratio of this approach will be high. Nevertheless, this is not an indication that the performance of the algorithm is high. Therefore, other widely used performance indicators were calculated. The accuracy of classification can be assessed by calculating the number of correctly identified values belonging to a class (true positives: TP), the number of correctly identified values not belonging to the class (true negatives: TN), and values that were either wrongly assigned to the class (false positives: FP) or not recognized as class instances (false negatives: FN) [75]. The values are derived from the binary confusion matrix. In the context of multi-class classification, the confusion matrix can be simplified to a binary form using the one vs. rest method, and TP, TN, FP, and FN values in this matrix can then be computed accordingly. Equations (4)–(7) were employed to calculate Recall, Specificity, Precision, and F1 Score based on these data.

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

$$Specificity = \frac{TN}{TN + FP} \tag{5}$$

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

$$F1\ Score = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \tag{7}$$

Recall, referred to as sensitivity or True Positive Rate, is a measure used to determine the proportion of true positive results in a classification algorithm. A high recall score indicates that the classifier accurately recognizes a majority of the positive cases. Specificity refers to the ratio of TN to all actual negatives. A high specificity score indicates that the classifier correctly detects negative cases. Precision is the proportion of TP out of the predicted positives. A high precision score indicates that the classifier accurately predicts positive instances. The F1 Score is a commonly used performance metric that effectively combines precision and recall. The performance metrics of the algorithms for each of the four classes and both groups are reported in Figures 5 and 6 to demonstrate their testing capabilities.

The results indicate that the models functioned adequately. Based on the average performance scores, DTC and KNN performed most effectively in PT.01. They were followed by SVC, RFC, NBC, and LRC in that order. In the PT.02 group, DTC and NBC achieved the highest results, followed by RFC, SVC, KNN, and LRC.

Subsequently, the models were evaluated utilizing validation techniques to obtain comprehensive performance indices, as determined by confusion matrices. Resampling is a widely used method in ML for evaluating and selecting models. Various resampling methods include bootstrapping, k-fold cross-validation, leave-one-out cross-validation, and repeated holdout validation. This study employed repeated random subsampling validation, also called "Monte Carlo cross-validation" or "shuffle-split cross-validation". This approach employs the holdout method iteratively k times, with different random seeds. The objective is to provide a more dependable performance estimate less influenced by how the data are divided into training and test sets [76]. Then, the average performance throughout these k iterations was computed. This method is beneficial for relatively small datasets. Compared with conventional k-fold cross-validation, the training and validation sets can be modified in size, and performance assessment is not excessively influenced by a certain division of data. Repeated random subsampling validation and k-fold cross-validation exhibit equivalent validation performance [77].

PT.01

LRC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.50	1.00	-	*0.63*
Specificity	0.70	1.00	0.87	1.00	*0.89*
Precision	0.70	1.00	0.50	-	*0.55*
F1 Score	0.82	0.67	0.67	-	*0.54*

KNN

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.83	1.00	-	*0.71*
Specificity	0.90	1.00	0.87	1.00	*0.94*
Precision	0.88	1.00	0.50	-	*0.59*
F1 Score	0.93	0.91	0.67	-	*0.63*

SVC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.67	1.00	-	*0.67*
Specificity	0.70	1.00	0.93	1.00	*0.91*
Precision	0.70	1.00	0.67	-	*0.59*
F1 Score	0.82	0.80	0.80	-	*0.61*

NBC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.50	1.00	-	*0.63*
Specificity	0.60	1.00	0.93	1.00	*0.88*
Precision	0.64	1.00	0.67	-	*0.58*
F1 Score	0.78	0.67	0.80	-	*0.56*

DTC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.83	1.00	-	*0.71*
Specificity	0.90	1.00	0.87	1.00	*0.94*
Precision	0.88	1.00	0.50	-	*0.59*
F1 Score	0.93	0.91	0.67	-	*0.63*

RFC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.67	1.00	-	*0.67*
Specificity	0.80	1.00	0.87	1.00	*0.92*
Precision	0.78	1.00	0.50	-	*0.57*
F1 Score	0.88	0.80	0.67	-	*0.59*

Figure 5. Evaluation of classifier algorithms based on performance indices for the PT.01 group.

PT.02

LRC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.57	0.50	1.00	*0.77*
Specificity	0.73	0.92	1.00	1.00	*0.91*
Precision	0.75	0.80	1.00	1.00	*0.89*
F1 Score	0.86	0.67	0.67	-	*0.55*

KNN

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	0.89	0.57	1.00	1.00	*0.87*
Specificity	0.73	0.92	1.00	1.00	*0.91*
Precision	0.73	0.80	1.00	1.00	*0.88*
F1 Score	0.80	0.67	1.00	-	*0.62*

SVC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	0.57	1.00	0.50	*0.77*
Specificity	0.64	1.00	1.00	1.00	*0.91*
Precision	0.69	1.00	1.00	1.00	*0.92*
F1 Score	0.82	0.73	1.00	-	*0.64*

NBC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	1.00	1.00	1.00	*1.00*
Specificity	1.00	1.00	1.00	1.00	*1.00*
Precision	1.00	1.00	1.00	1.00	*1.00*
F1 Score	1.00	1.00	1.00	-	*0.75*

DTC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	1.00	1.00	1.00	*1.00*
Specificity	1.00	1.00	1.00	1.00	*1.00*
Precision	1.00	1.00	1.00	1.00	*1.00*
F1 Score	1.00	1.00	1.00	-	*0.75*

RFC

Performance Index	Classes 0	1	2	3	*Ave.*
Recall	1.00	1.00	0.50	0.50	*0.75*
Specificity	1.00	0.85	1.00	1.00	*0.96*
Precision	1.00	0.78	1.00	1.00	*0.94*
F1 Score	1.00	0.88	0.67	-	*0.64*

Figure 6. Evaluation of classifier algorithms based on performance indices for the PT.02 group.

Table 6 presents the overall accuracy of the cross-validation. Confusion matrices for randomly selecting states were calculated by repeating the process ten times for the PT.01 and PT.02 groups.

Table 6. Overall accuracy values for 10 random states.

PT.01												
Algorithms	**ACC**										**Ave.**	**Rank**
	RS = 1	**RS = 2**	**RS = 3**	**RS = 4**	**RS = 5**	**RS = 6**	**RS = 7**	**RS = 8**	**RS = 9**	**RS = 10**		
LRC	0.71	0.82	0.65	0.53	0.76	0.71	0.88	0.65	0.71	0.82	0.72	6
KNN	0.82	0.82	1.00	0.82	0.94	1.00	1.00	0.82	0.88	0.88	0.90	2
SVC	0.76	0.76	0.76	0.71	0.88	0.71	0.88	0.71	0.82	0.88	0.79	5
NBC	0.71	1.00	0.82	0.71	0.76	0.94	0.76	0.88	0.76	0.94	0.83	4
DTC	0.82	1.00	1.00	0.82	0.94	1.00	1.00	1.00	0.88	0.94	0.94	1
RFC	0.76	1.00	1.00	0.71	0.82	1.00	1.00	0.94	0.76	0.88	0.89	3
PT.02												
Algorithms	**ACC**										**Ave.**	**Rank**
	RS = 1	**RS = 2**	**RS = 3**	**RS = 4**	**RS = 5**	**RS = 6**	**RS = 7**	**RS = 8**	**RS = 9**	**RS = 10**		
LRC	0.80	0.90	0.65	0.55	0.65	0.75	0.85	0.85	0.75	0.85	0.76	5
KNN	0.80	0.95	0.80	0.75	0.85	0.75	0.60	0.90	0.80	0.90	0.81	4
SVC	0.80	0.95	0.65	0.45	0.55	0.65	0.55	0.85	0.75	0.80	0.70	6
NBC	1.00	0.85	0.80	0.90	0.90	0.85	0.95	0.85	0.85	0.95	0.89	3
DTC	1.00	1.00	0.80	1.00	0.95	1.00	0.95	1.00	0.95	0.95	0.96	1
RFC	0.90	1.00	0.85	0.85	0.90	0.95	0.95	0.80	0.90	0.90	0.90	2

Based on the validation groups, LRC and SVC did not achieve an accuracy of 0.80 for PT.01 and PT.02. However, DTC demonstrated higher accuracy performance for both groups, followed by KNN and RFC in PT.01, and RFC and NBC in PT.02.

Subsequently, the cross-validation performance indices were computed by repeating the procedure ten times to generate confusion matrices and performance indices for each class within the PT.01 and PT.02 groups. The average scores of performance indicators for four classes are computed and presented in Table 7 for each classifier.

Table 7. Summary of average performance indices for classifiers from cross-validation.

PT.01					
Algorithms	**Recall (avg)**	**Specificity (avg)**	**Precision (avg)**	**F1 Score (avg)**	**Ave.**
KNN	0.66	0.96	0.65	0.64	0.73
NBC	0.61	0.94	0.61	0.58	0.68
DTC	0.71	0.98	0.70	0.69	0.77
RFC	0.63	0.96	0.61	0.60	0.70
PT.02					
Algorithms	**Recall (avg)**	**Specificity (avg)**	**Precision (avg)**	**F1 Score (avg)**	**Ave.**
KNN	0.80	0.91	0.82	0.80	0.83
NBC	0.78	0.96	0.72	0.72	0.79
DTC	0.91	0.99	0.92	0.91	0.93
RFC	0.72	0.95	0.77	0.72	0.79

The DTC demonstrates outstanding predictive performance in both groups when evaluating performance indices. In PT.01, KNN and RFC both show strong performance, while in PT.02, KNN, NBC, and RFC exhibit notable performance.

The DTC method demonstrates superior results for both groups. To enhance the algorithm's performance, the parameters of the algorithm were adjusted. Grid Search is an optimization technique that involves systematically testing different parameter values for an algorithm to fine-tune its performance for a given data set. It is known for its efficiency in terms of computational time and resource usage [78–80]. The algorithm parameters of this classifier consist of "criterion" and "splitter". The "criterion" refers to the function used to assess the quality of a split, while "splitter" denotes the approach employed to select the split at each node. The options for "criterion" and "splitter" are {"gini", "entropy", "log_loss"} and {"best", "random"}, respectively. All combinations for ten random states were obtained by ten times run, and the average accuracy scores of these states for PT.01 and PT.02 are presented in Figure 7.

0.9233 in PT.01 and 0.9708 in PT.02 group are the superior score outcomes of the parameter combinations. The optimal combination for PT.01 was "criterion" = "entropy"

and "splitter" = "best", but for the PT.02 group, "splitter" = "best" gave superior results with all criterion alternatives.

PT.01		splitter	
		best	random
criterion	gini	0.9208	0.8842
	entropy	**0.9233**	0.9000
	log_loss	0.9150	0.8783

PT.02		splitter	
		best	random
criterion	gini	**0.9708**	0.9283
	entropy	**0.9708**	0.9083
	log_loss	**0.9708**	0.9342

Figure 7. Average accuracy score matrices of grid search. The bold numbers indicate the superior results.

4.5. Benefits and Limitations

The primary benefit of this approach is its ease of application in the early stages of projects. This is achieved by utilizing data on the TC and total risk score of previously completed projects. Furthermore, it can be deployed or modified to projects during the execution phase at any time by utilizing the updated cost and total risk score data. Moreover, the suggested approach covered the financial implications of risks, which are excluded from cost estimations and mainly assessed in risk registers. Additionally, the proposed approach is simple to use because it employs total cost and risk scores rather than their breakdowns. Another advantage of this approach is that the cost overrun ratio classes offer a clear comprehension of the gravity of cost escalations. Also, executives may develop a mitigating action plan based on the expected percentage increase in TC.

On the other hand, certain limitations might be considered that have potential for improvement within the framework of the proposed approach. Firstly, there are risk perception characteristics that might influence the calculation of total risk score from one project to another and cost overrun ratio classes. Similarly, each company can assess the risk probability and impact scales variably, influenced by project type and organizational dynamics, and may use different ML algorithms based on the nature of their data. Furthermore, every project generates an equivalent amount of data as it updates its risks and costs. Hence, there might be a constraint on the quantity of data available. Moreover, it is known that receiving updates on the risk and cost of a construction project during the execution phase is more challenging compared with obtaining data regarding the tender cost or completion cost at the end of the project. Ultimately, the DTC algorithm, which is the best-performing algorithm on the data in this study, has a tendency to be biased towards dominant features, which might result in overfitting the training dataset. Therefore, the selection of algorithms and parameters should be contingent on the specific characteristics of the data. These limitations may diverge the results among different companies and alternative ML algorithms.

5. Conclusions

Cost estimation plays a crucial role in the construction industry. Over the years, numerous researchers have focused on this subject and assessed the total completion cost or cost overrun ratio using various methodologies. However, there is a need for cost estimation models during the execution phase of construction projects. During the execution phase, the intense workload limits the time available for cost estimation compared with earlier phases. Consequently, it is essential to deliver estimations promptly and efficiently throughout this period. Furthermore, contract terms or changes in the scope of work and associated risks due to site conditions are primarily incorporated into cost estimation processes after identification. However, it is challenging to periodically integrate all detected risks into the completion cost. Utilizing risk scores reported in monthly risk registers presents an opportunity to apply these scores to the completion cost as a cost overrun ratio. A swift and user-friendly approach was proposed to address this gap. This approach suggests utilizing ML classification algorithms to forecast the class of cost overrun ratios based on total risk scores. LRC, KNN, SVC, NBC, DTC, and RFC were employed owing to their widespread application, ease of use, and capacity to evaluate performance. The efficiencies

of the algorithms were evaluated using accuracy and additional performance metrics such as recall, specificity, precision, F1 Score, and resampling techniques. The proposed approach was tested using 110 data points sourced from a globally operating construction firm. Data from eleven projects were segregated into two groups, and six proposed ML classification algorithms were employed. DTC, KNN, and RFC in the PT.01 group and DTC, KNN, NBC, and RFC in the PT.02 group demonstrated superior performance based on performance indices. After evaluating models with repeated random subsampling validation, the DTC algorithm yielded the most favorable outcomes in terms of overall accuracy and performance indices for both groups. Grid Search fine-tuned the parameters of this algorithm. The optimal combinations in DTC were determined as entropy-best in PT.01 and gini/entropy/log_loss-best in PT.02.

Although the proposed approach has some limitations, including risk perception about evaluating risks, data shortages or restrictions to gather, and selection of the proper ML algorithm depending on data characteristics, the suggested approach offers benefits in terms of its user-friendly implementation and the prompt and meaningful outcomes it provides to project executives at any stage of a construction project. Additionally, this study facilitates future investigations and analyses in the construction field. The authors continue to conduct research on and investigations of these subjects.

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